Recap: A notion of discontinuous functions.

Limits of functions: $\lim_{x \to +\infty} (f(x))$

$\lim_{x \to -\infty} (f(x))$

$\lim_{x \to c} (f(x))$

Limits and cts functions are related $\lim_{x \to c} (f(x)) = f(c)$.

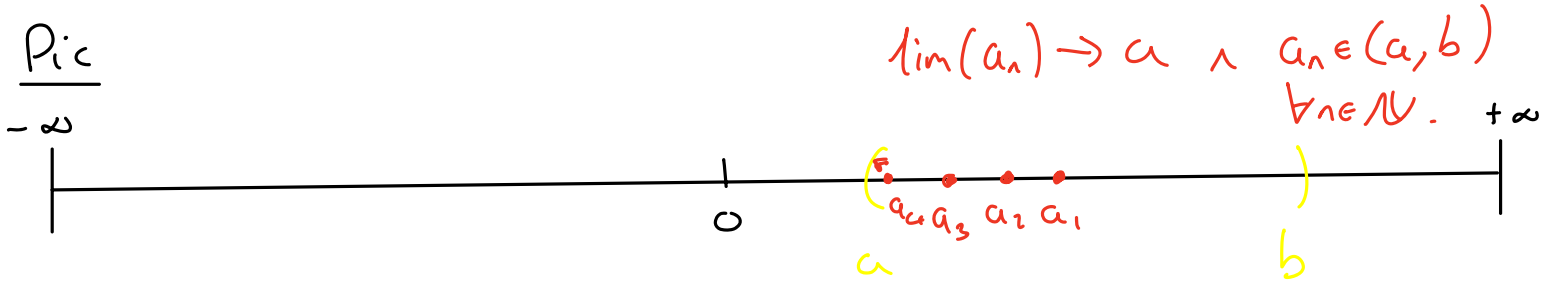
Split up limits into left and right sided limits.

$\lim_{x \to c^-} (f(x))$ $\lim_{x \to c^+} (f(x))$

7 - Sequential Continuity

Theorem 7.1: (Sequences and limit points)

Pic



A point $x \in \mathbb{R}$ is a limit point of a set $D \subset \mathbb{R}$ iff there exists a sequence (a_n) with $\lim(a_n) = x$ and $a_n \in D - \{x\}$ for each $n \in \mathbb{N}$.

Proof: Exercise.

Theorem 7.2: Sequential Criterion of Functional Limits

Let $f: D \to \mathbb{R}$ and $c \in \overline{D}$. Then $\lim_{x \to c}(f(x)) = L$ iff for every sequence (a_n) with $\lim(a_n) = c$ and $a_n \in D \setminus \{c\}$ satisfies $\lim(f(a_n)) = L$.

Proof: ($\Rightarrow$) Suppose $\lim_{x \to c}(f(x)) = L$. Then for all $\varepsilon > 0$ there exists a $\delta > 0$ s.t

$$|f(x) - L| < \varepsilon$$

whenever $0 < |x - c| < \delta$ and $x \in D$. By the above theorem, such a sequence (a_n) exists with the required properties. Hence, there exists $N \in \mathbb{N}$ s.t

$$0 < |a_n - c| < \delta$$

for every $n > N$. Hence,

$$|f(a_n) - L| < \varepsilon$$

for all $n > N$.

($\Leftarrow$) Suppose towards a contradiction that the sequence (a_n) satisfies the required properties, and $\lim(f(a_n)) = L$ but $\lim_{x \to c}(f(x)) \neq L$. Then there exists an $\varepsilon > 0$ s.t for every $\delta > 0$, there exists $x \in D$ with $0 < |x - c| < \delta$ and $|f(x) - L| \geq \varepsilon$.

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in D)(0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon)$$

$$(\exists \varepsilon > 0)(\forall \delta > 0)(\exists x \in D)(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon)$$

Hence, for every $n \in \mathbb{N}$ there exists $a_n \in D \setminus \{c\}$ with $|a_n - c| < 1/n$ and $|f(a_n) - L| \geq \varepsilon$.

It follows that $\lim(a_n) = c$, but $|f(a_n) - L| \geq \varepsilon$ $\forall n \in \mathbb{N}$. This is a contradiction since $\lim(f(a_n)) = L$.

□

Corollary 7.3 : Divergence Criterion for Functional Limits

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$. Then $\lim_{x \to c}(f(x))$ does not exist if there exists two distinct sequences (a_n) and

(b_n), with $\lim(a_n) = \lim(b_n) = c$ and $a_n, b_n \in D \setminus \{c\}$ $\forall n \in \mathbb{N}$, such that $\lim(f(a_n)) \neq \lim(f(b_n))$.

Ex: The Dirichlet function $f: \mathbb{R} \to \mathbb{R}$ is given by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}.$$

Then f is not cts anywhere.

Proof: Let $c \in \mathbb{R}$. Clearly, there exists $a, b \in \mathbb{R}$ with $c \in (a, b)$. Since, $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists an $a_1 \in \mathbb{Q}$ with $a_1 \in (a, c)$. Similarly, there exists $a_2 \in \mathbb{Q}$ with $a_2 \in (a_1, c)$. Continuing this defines a monotone increasing sequence (a_n) with $a_n \in \mathbb{Q}$ $\forall n \in \mathbb{N}$ and $\lim(a_n) = c$. Hence, $\lim(f(a_n)) = 1$. By a similar argument there exists a sequence (b_n) with $b_n \in \mathbb{R} \setminus \mathbb{Q}$ $\forall n \in \mathbb{N}$ and $\lim(b_n) = c$. Hence, $\lim(f(b_n)) = 0$.

It follows that $f(x)$ is discontinuous for all $x \in \mathbb{R}$.

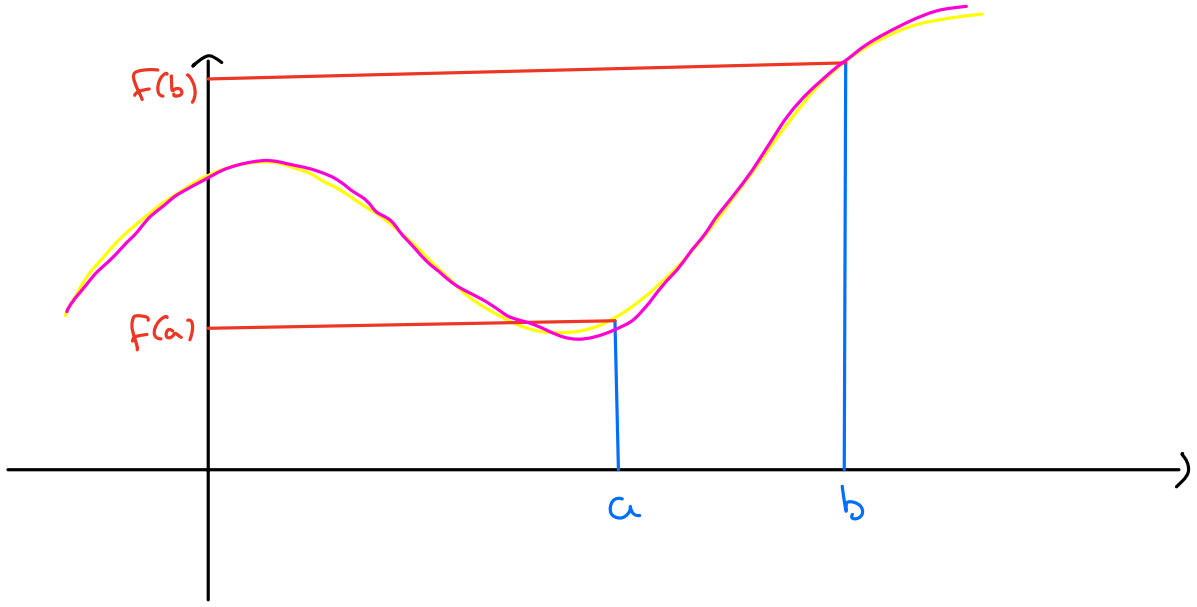
8 - Intermediate and Extreme Value Theorem

Theorem 8.1 : Boundedness Theorem

Let $f: [a,b] \to \mathbb{R}$ be a cts on $[a,b]$.

Then the function is bounded.

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Theorem 8.2 : Extreme Value Theorem

Let $f: [a,b] \to \mathbb{R}$ be a cts function on $[a,b]$.

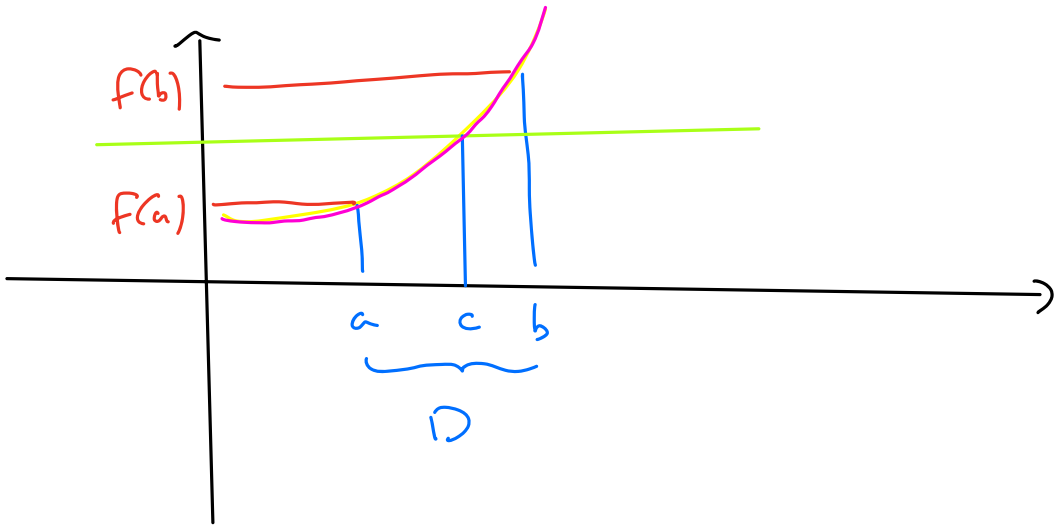
Then f attains its bounds.

Theorem 8.3 : Intermediate Value Theorem

Let $f: D \to \mathbb{R}$ be a cts on D. Then for every $[a,b] \subset D$, and $\alpha \in \mathbb{R}$, with $f(a) < \alpha < f(b)$

or $f(b) < \alpha < f(a)$, there exists $c \in (a,b)$ such that $f(c) = \alpha$.

Pic:



Ex: Let $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

where $a_i \in \mathbb{R}$ for each $i \in \{0, 1, 2, 3\}$. Then p has at least one real root.

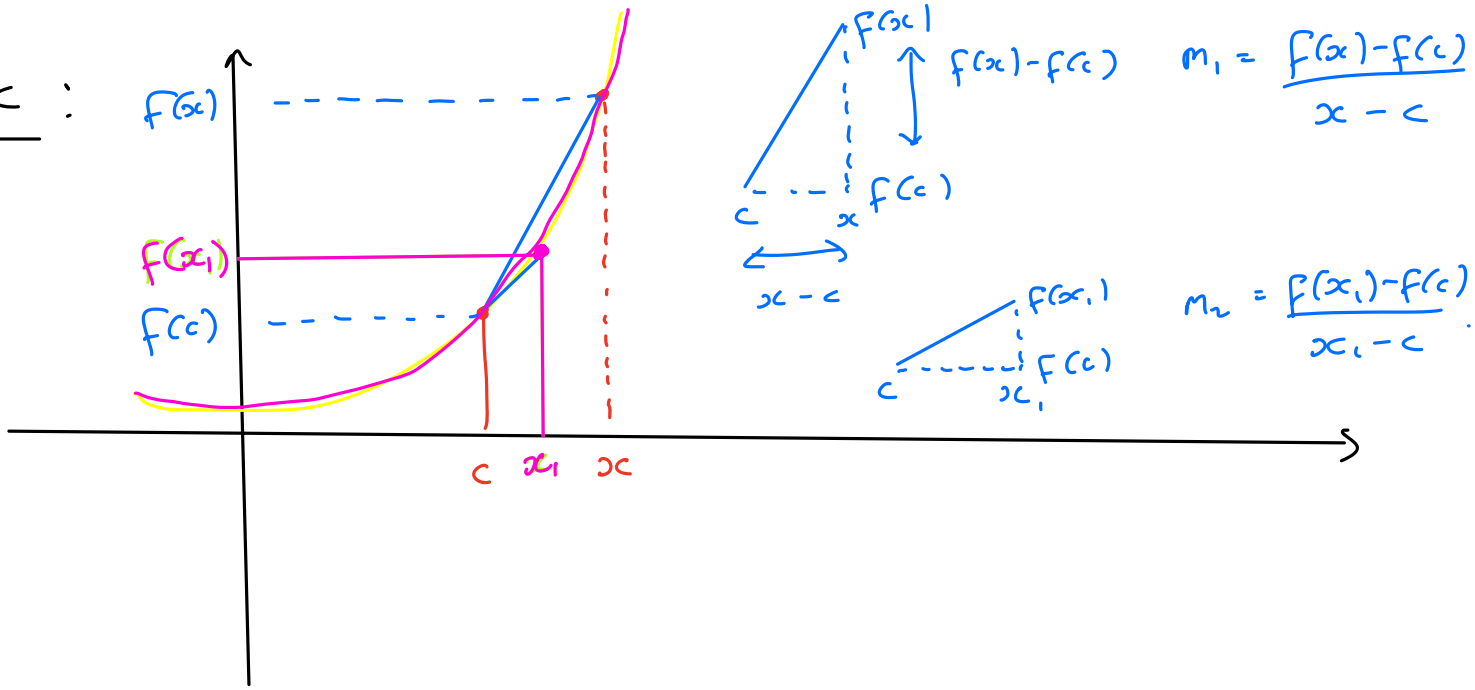
Proof: Since, $\lim_{x \to +\infty} (p(x)) = +\infty$, and

$\lim_{x \to -\infty} (p(x)) = -\infty$. It follows that there exists $a \in \mathbb{R}$ and $b \in \mathbb{R}$ such that $p(x) < 0$ for $x \leq a$ and $p(x) > 0$ for $x \geq b$. Since p is cts on $\mathbb{R}$ it follows by the IVT (intermediate value theorem) that

$p|_{[a,b]} : [a,b] \to \mathbb{R}$ is cts on $[a,b]$. Hence, there exists $c \in (a,b)$ s.t $p(c) = 0$. That is p has at least one real root. ∎

Chapter 5 : Differentiation and Power Series

Pic :



Let $f: D \to \mathbb{R}$. Then f is said to be differentiable at the point $c \in D$ if the limit

$$f'(c) := \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. The number $f'(c)$ is called the derivative of f at c.

Remark: The defn can be equivalently written as

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}.$$

Ex: $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = 1$.

Let $c \in \mathbb{R}$. Then

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0} \frac{1 - 1}{h} = 0.$$

Hence, $f'(c)$ exists $\forall c \in \mathbb{R}$ and $f'(c) = 0$.

iii) $r: \mathbb{R} \to \mathbb{R}$ given by $r(x) = x^2$. Let $c \in \mathbb{R}$.

Then

$$r'(c) = \lim_{h \to 0} \left(\frac{r(c+h) - r(c)}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{c^2 + 2ch + h^2 - c^2}{h} \right)$$

$$= \lim_{h \to 0} \left(2c + h \right)$$

$= 2c$.

Hence, $r'(x) = 2x$.